

FUKUSHIMA, Japan

WMO: 475950

Lat: 37.7584N

Lon: 140.4710E

Elev: 69

StdP: 100.50

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.8	-10.1	1.6	3.1	-8.9	1.8	2.2	8.3	2.7	7.5	2.9	1.8	270	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	34.2	24.9	32.7	24.4	31.1	23.5	25.7	31.9	25.2	30.8	24.6	29.6	3.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.2	28.5	23.6	18.5	28.0	23.0	17.9	27.6	80.0	32.4	77.5	31.0	75.0	29.7	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.5	5.9	-5.7	36.5	1.0	1.0	-6.4	37.2	-7.0	37.8	-7.6	38.4	-8.3	39.1		
(5)			-6.7	26.9	0.9	0.6	-7.3	27.3	-7.9	27.7	-8.4	28.0	-9.1	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	2.0	2.7	6.1	12.0	17.6	21.1	25.0	26.3	22.1	16.1	9.7	4.6
	DBStd	8.96	2.23	2.63	3.26	3.72	3.22	2.90	3.31	3.16	3.36	3.03	3.18	2.71
	HDD10.0	817	249	203	129	22	1	0	0	0	0	1	44	169
	HDD18.3	2349	507	436	379	192	52	9	1	0	6	83	258	426
	CDD10.0	2217	0	1	8	81	236	332	466	505	362	189	36	1
	CDD18.3	708	0	0	0	2	29	91	208	247	118	13	0	0
	CDH23.3	5643	0	0	0	34	245	559	1759	2266	737	42	0	0
	CDH26.7	2058	0	0	0	6	54	147	694	943	212	3	0	0
Wind	WSAvg	2.3	2.5	2.7	2.8	2.8	2.6	2.2	2.1	2.1	1.9	2.0	2.2	2.4
Precipitation	PrecAvg	1216	64	51	74	78	86	124	171	153	155	120	74	67
	PrecMax	1569	129	101	148	144	145	211	315	527	296	358	143	167
	PrecMin	843	26	14	10	3	21	63	47	30	14	19	9	24
	PrecStd	174	30	21	37	42	36	48	72	106	82	89	34	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	14.6	20.7	27.0	30.8	32.1	35.6	36.2	33.3	26.7	20.5	15.1
		MCWB	7.0	8.2	10.8	14.8	18.2	21.2	25.1	25.3	24.1	19.9	14.0	10.5
	2%	DB	8.9	11.5	17.2	23.6	28.0	30.1	34.0	34.7	31.0	24.4	18.5	12.6
		MCWB	4.6	6.5	9.3	13.4	16.6	20.3	25.0	25.1	23.4	18.2	13.0	8.3
	5%	DB	7.2	9.3	14.6	21.2	26.1	28.4	32.3	33.3	29.1	22.7	16.7	10.7
		MCWB	3.4	4.7	8.4	12.5	16.2	19.9	24.3	24.9	22.3	17.0	12.0	7.0
	10%	DB	5.9	7.4	12.2	19.0	24.1	26.7	30.6	31.7	27.3	21.2	15.2	9.2
		MCWB	2.6	3.5	7.1	11.4	15.5	19.4	23.7	24.4	21.6	16.2	11.3	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	9.7	12.7	18.1	20.5	23.6	26.5	26.7	25.2	21.4	15.9	11.1
		MCDB	10.5	14.1	17.7	22.5	26.5	28.4	33.3	33.1	30.4	25.1	18.6	13.7
	2%	WB	5.2	6.8	10.7	15.2	19.0	22.5	25.6	26.0	24.3	19.5	14.3	9.0
		MCDB	8.1	10.5	15.2	21.1	24.5	27.1	31.9	32.2	29.0	22.6	16.8	12.0
	5%	WB	3.9	5.2	9.1	13.7	17.8	21.5	25.0	25.5	23.6	18.2	13.0	7.3
		MCDB	6.5	8.7	13.8	18.6	23.0	25.9	30.8	31.3	27.7	21.2	16.0	10.4
	10%	WB	2.8	3.8	7.3	12.5	16.8	20.5	24.4	24.9	22.8	17.1	11.6	6.0
		MCDB	5.6	7.1	11.7	17.4	21.8	24.8	29.3	30.3	26.4	20.3	14.5	8.9
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.6	9.1	10.4	10.3	8.6	7.7	7.9	7.6	8.2	8.5	7.1
		MCDBR	9.1	11.0	13.9	14.7	14.3	12.0	10.7	10.3	9.7	9.9	10.9	9.8
		MCWBR	5.9	6.9	8.0	7.0	5.8	4.7	3.8	3.4	3.9	5.0	6.5	6.4
	5% WB	MCDBR	8.6	10.6	12.5	11.7	11.0	9.2	9.5	9.4	8.4	7.7	9.6	9.5
		MCWBR	6.0	7.1	8.2	6.6	5.6	4.2	3.8	3.4	3.8	4.7	6.6	6.7
Clear-Sky Solar Irradiance	taub	0.426	0.489	0.578	0.636	0.654	0.664	0.670	0.659	0.563	0.504	0.456	0.428	
	taud	1.915	1.775	1.666	1.615	1.620	1.657	1.697	1.716	1.890	1.952	1.991	1.982	
	Ebn at Noon	702	705	685	678	677	670	661	654	693	689	669	668	
	Edh at Noon	142	187	229	255	258	248	237	228	181	154	132	125	
All-Sky Solar Radiation	RadAvg	2.07	2.81	3.72	4.58	5.11	4.81	4.41	4.44	3.63	2.93	2.18	1.77	
	RadStd	0.14	0.18	0.25	0.44	0.52	0.33	0.69	0.55	0.29	0.29	0.14	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.52	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page